

The brain of drones miniaturised

Engineers at Massachusetts Institute of Technology (MIT) have designed a computer chip that uses a fraction of the power of larger drone computers and is tailored for drones as small as bottlecaps. The team, led by Sertac Karaman and Vivienne Sze, developed a low-power algorithm,

in tandem with pared-down hardware, to create the specialised computer chip.

"This new approach to programming robots, which involves thinking about hardware and algorithms jointly, is key to scaling them down," Karaman said.

The new chip processes streaming images at 20 frames per second and automatically carries out commands to adjust a drone's orientation in space. The streamlined chip performs all these computations while using just below two watts of power—making it much more efficient than current drone-embedded chips.

According to Karaman, the team's design is the first step toward engineering "the smallest intelligent drone that can fly on its own." He ultimately envisions disaster-response and search-and-rescue missions in which insect-sized drones flit in and out of tight spaces to examine a collapsed structure or look for trapped individuals.



The computer chip designed by MIT engineers is tailored for a drone as small as a bottlecap (Image courtesy: Christine Daniloff/MIT)

Machine learning to detect heart arrhythmias

Stanford University researchers have shown that a machine-learning model can identify heart arrhythmias from an electrocardiogram (ECG) better than an expert doctor. The automated approach is also significant because it could make quality care readily accessible in areas where resources are scarce.

The team, led by professor Andrew Ng, trained their deep-learning algorithm on data collected from iRhythm's wearable ECG monitor. They took approximately 30,000, 30-second clips from various patients that represented a variety of arrhythmias. To test the accuracy of the algorithm, the researchers gave a group of three expert cardiologists 300 undiagnosed

clips and asked them to reach a consensus about any arrhythmias present in the recordings. Working with these annotated clips, the algorithm could then predict how those cardiologists would label every second of other ECGs with which they were presented, in essence, giving a diagnosis.

The group had six different cardiologists, working individually, diagnose the same 300-clip set. The researchers then compared which more closely matched the consensus opinion—the algorithm or the cardiologists working independently. They found that the algorithm is competitive with the cardiologists, and able to outperform cardiologists on most arrhythmias.